

SMARTBRIDGE PROJECT DOCUMENTATION

Project Description

Emotion Detection & Learning Support Engine is an AI-powered educational assistant that analyzes student text, detects emotions using Machine Learning and NLP, and generates personalized learning guidance using Google Gemini AI. It improves engagement, reduces frustration, and provides adaptive learning support.

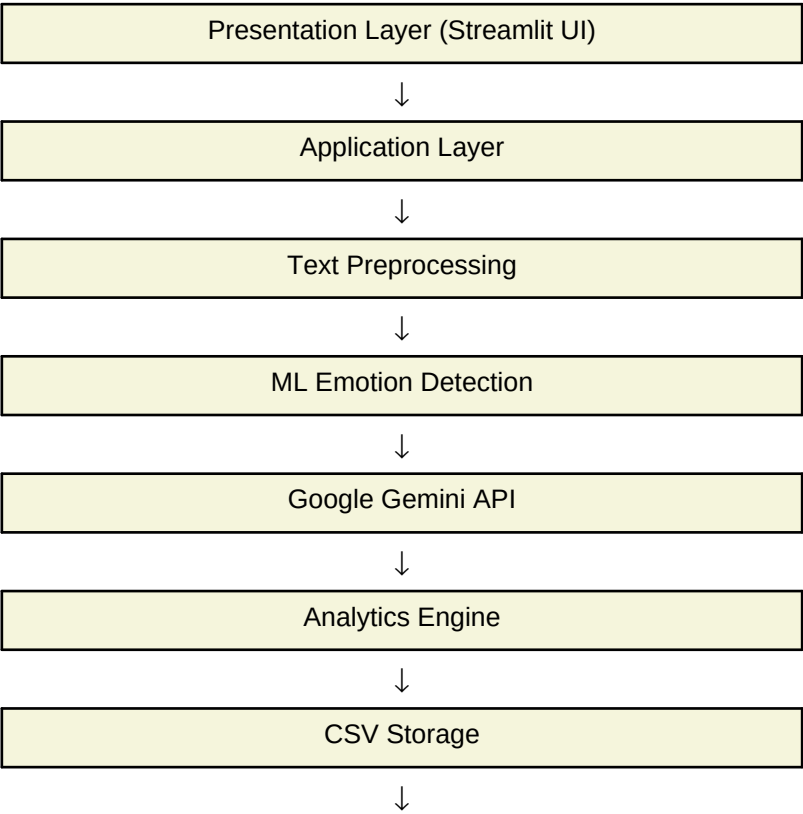
Scenarios

Scenario 1	Student enters academic text. System detects emotion (Confused, Frustrated, Curious, Confident) and generates Gemini guidance.
Scenario 2	Personalized study recommendations, motivation and concept explanations are generated.
Scenario 3	Emotion analytics dashboard tracks emotional learning progress.

Technical Architecture

Streamlit UI → Python Backend → Text Preprocessing → NLP Pipeline → ML Emotion Detection
→ Google Gemini API → Analytics → CSV Storage → Plotly Dashboard

Architecture Diagram



ER Diagram

Student → Emotion History → Emotion Prediction → AI Response → Learning Recommendation → Analytics Record

Pre-requisites

Python 3.10+
Streamlit
Pandas
NumPy
Scikit-Learn
Plotly
Google Gemini API
Git
GitHub
VS Code

Project Workflow	
Milestone	Activities
Milestone 1	Problem Identification, Requirement Analysis, Environment Setup
Milestone 2	Data Collection, Dataset Preparation, Text Cleaning, Feature Engineering
Milestone 3	Emotion Detection Model Development, Model Training, Evaluation, Accuracy Testing
Milestone 4	Frontend Development, Streamlit UI, Dashboard Design, Gemini Integration
Milestone 5	Testing, Bug Fixing, Performance Optimization
Milestone 6	Deployment, Documentation, Final Presentation
Website / Application Pages	
Home	Landing page introducing Emotion Detection & Learning Support Engine. [Screenshot Placeholder]
About	Objectives, technologies, benefits, workflow. [Screenshot Placeholder]
Emotion Detection	Students enter text, predicted emotion and Gemini guidance displayed. [Screenshot Placeholder]
Analytics	Emotion charts, session history, learning statistics. [Screenshot Placeholder]
History	Previous interactions, emotions, AI responses, date/time. [Screenshot Placeholder]

Conclusion

Emotion Detection & Learning Support Engine combines Machine Learning, NLP, and Generative AI to provide personalized educational support. Future enhancements include multilingual support, voice emotion recognition, cloud deployment, and adaptive learning recommendations.